

## CS2206: Computer Architecture-2026

### Lab 7: MIPS RISC processor Single Cycle design

*Complete MIPS design with control unit ( read chapter 7 of the test book for details) and processor should support( lw, sw, add, sub, or , addi, and , beq, slt, jump, jr, and jal instructions) and demonstrate with the test program given. (70 points)*

*(b) Create two new test program examples to demonstrate that your design is working ( For Eg. sum of n umbers , find maximum of a set of numbers in that is stored in memory , include instructions to initialize memory in all cases) etc. (30 points)*

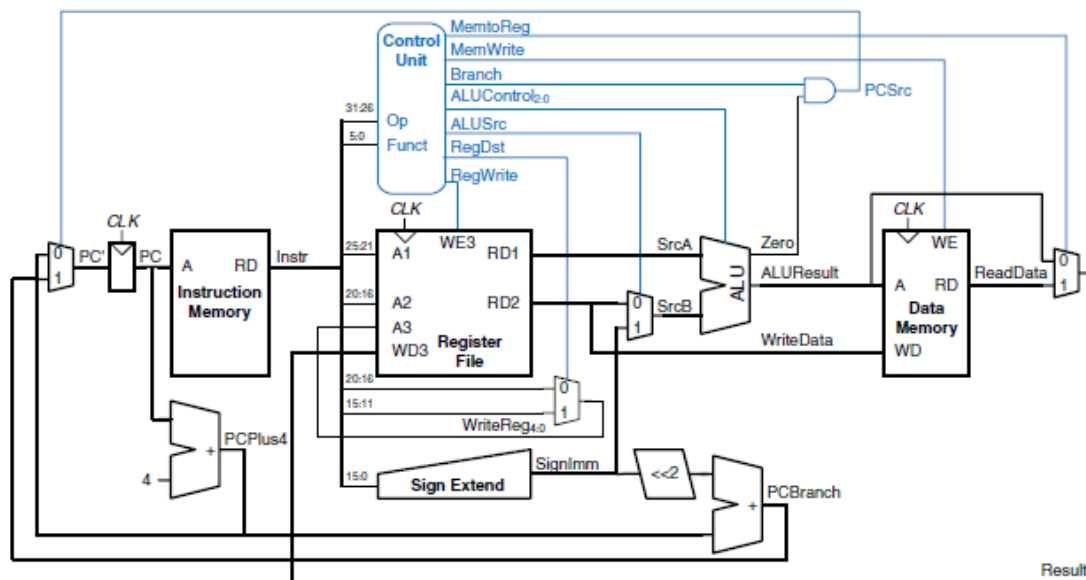
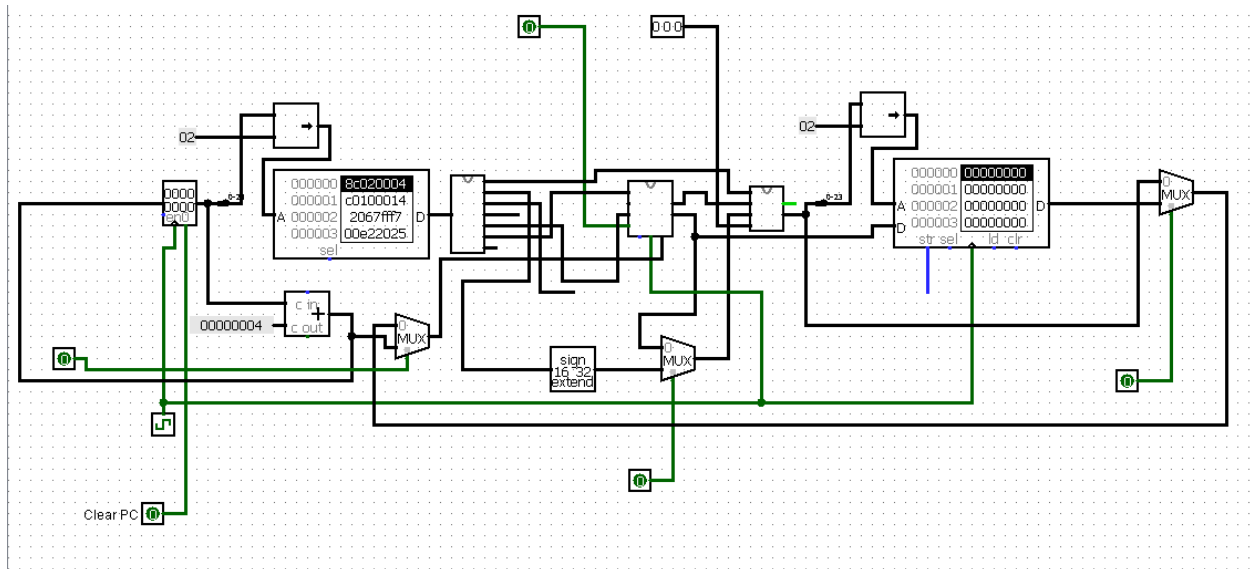
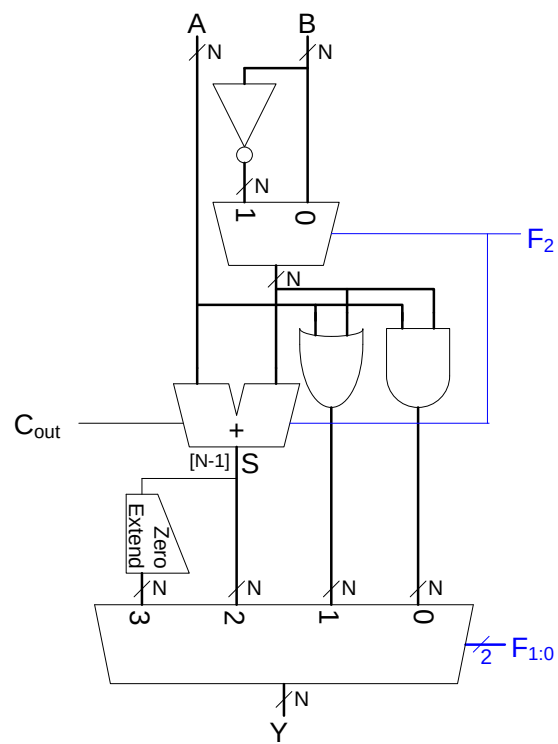


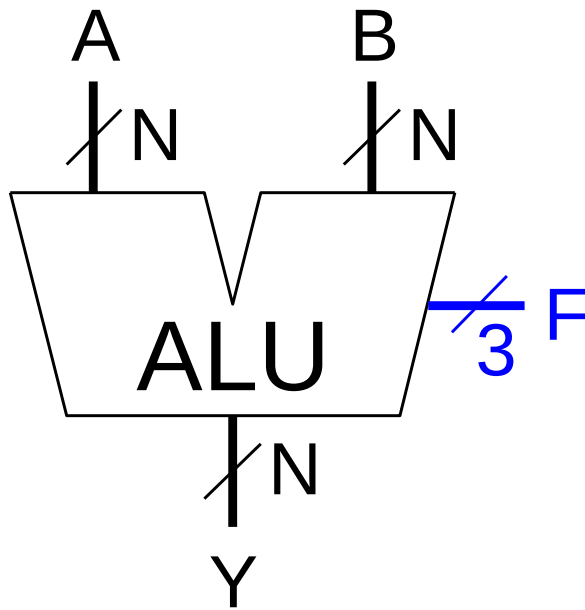
Figure 7.11 Complete single-cycle MIPS processor



***Demonstrate at end of the lab and Submit single circ file( test programs) with name yourRollNo\_Lab7.circ to***

## Appendix A





| $F_{2:0}$ | Function        |
|-----------|-----------------|
| 000       | $A \& B$        |
| 001       | $A \mid B$      |
| 010       | $A + B$         |
| 011       | not used        |
| 100       | $A \& \sim B$   |
| 101       | $A \mid \sim B$ |
| 110       | $A - B$         |
| 111       | SLT             |